

## Logistic Regression Hyperparameters — Short Notes

Hyperparameters are **settings we choose before training** the Logistic Regression model.

### 1. penalty

Controls **regularization** to reduce overfitting.

- $l_1$  → Lasso → can make coefficients **zero**
  - $l_2$  → Ridge → shrinks coefficients
  - elasticnet →  $L_1 + L_2$
  - None → no regularization
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### 2. C

Controls **regularization strength**.

Remember:

**C is inverse of regularization strength.**

Small C → Strong regularization → Simpler model

Large C → Weak regularization → More complex model

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### 3. solver

The **algorithm used to find the best model parameters**.

Common ones:

- lbfgs → good default
  - liblinear → small datasets
  - saga → large datasets / supports  $L_1$  & ElasticNet
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### 4. max\_iter

Maximum number of **iterations** the solver can perform.

`LogisticRegression(max_iter=1000)`

Increase it if you get a **convergence warning**.

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### 5. class\_weight

Useful for **imbalanced datasets**.

`LogisticRegression(class_weight='balanced')`

It gives more importance to the minority class.

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## 6. multi\_class

Controls how Logistic Regression handles **multiple classes**.

Common approaches:

- ovr → One-vs-Rest
  - multinomial → considers all classes together
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## 7. l1\_ratio

Used with **Elastic Net**.

Controls the balance between L1 and L2:

$l1\_ratio = 1 \rightarrow L1$

$l1\_ratio = 0 \rightarrow L2$

$l1\_ratio = 0.5 \rightarrow 50\% L1 + 50\% L2$



**Easy Remember**

**Penalty** → What regularization?

**C** → How strong?

**Solver** → How to optimize?

**max\_iter** → How many iterations?

**class\_weight** → Handle imbalance

**multi\_class** → Handle multiple classes

**l1\_ratio** → L1 vs L2 balance